

Summary

Ph.D. candidate in Industrial & Systems Engineering with 5 years of industry experience building production-scale software and optimization solutions. Specializing in fair machine learning, federated learning, and privacy-enhancing techniques using synthetic data and data distillation. Seeking a Summer 2026 Applied Scientist internship in Machine Learning, leveraging expertise in optimization to solve real-world challenges.

Education

Lehigh University, Bethlehem, PA	
Ph.D. in Industrial and Systems Engineering	08/2022 – Present
National Chiao Tung University, Hsinchu, Taiwan	
M.S. in Industrial Engineering and Management	09/2015 – 06/2017
B.B.A. in Management Science	09/2011 – 06/2015

Technical Skills

- **Programming Languages & Database:** Python, C#, PL/SQL, T-SQL, Matlab, MySQL, SQL Server, Oracle
- **ML, AI & Optimization:** PyTorch, TensorFlow, Keras, scikit-learn, CPLEX, Gurobi, GLPK, AMPL
- **Libraries & Toolkits:** Pandas, Numpy, ASP.NET, VSTO, Git, CI/CD

Work Experience

Advanced Micro Devices, Inc. (AMD), Hsinchu, Taiwan	04/2021 – 07/2022
Software System Designer (AsiaOps Planning - TW Team)	
• Engineered and optimized a unified data feed solution using MSSQL, C#, and VSTO to automate SCM and S&OP processes, reducing user workload by 50% and eliminating manual scheduling • Customized and enhanced Kinaxis RapidResponse functionalities via JIRA to improve supply chain planning modules and decision-making automation, reducing user support load and streamlining planning workflows • Delivered scalable integration of GPU/CPU planners by collaborating with cross-functional teams, managing concurrent tasks across supply chain modules, improving coordination efficiency and execution speed	
AU Optronics Corp. (AUO), Hsinchu, Taiwan	
Senior Software Engineer (Production Planning Information Team)	09/2017 – 04/2021
• Led a 3-member cross-functional team to develop MIP-based scheduling and resource optimization systems using IBM CPLEX, PL/SQL, and C#, reducing operational time by 90% and replacing legacy rule-based heuristics • Built and deployed an end-to-end optimization pipeline integrated with SQL Server and Oracle DB, boosting scheduling precision, reducing production delays, and increasing order success rate by 2X while cutting buyer workload by 95% • Designed interactive dashboards and KPI tracking tools with VSTO and ASP.NET, enabling real-time decision-making and streamlined daily/weekly production monitoring	

Selected Projects

Synthetic Data for Mitigating Unfairness in Collaborative Machine Learning	06/2024 – 07/2025
LU Research Assistant	[Paper] [Code]
• Formulated a novel bilevel optimization model for distributed learning, integrating data distillation and unfairness mitigation to generate representative synthetic datasets, ensuring fairness, robustness, and privacy in server-side training • Implemented a single-round machine learning framework with synthetic data generation and differential privacy to reduce client-server communication and enable scalable collaborative learning using Python, PyTorch, and MySQL	
ATP Pegging Optimization	09/2020 – 03/2021
• Designed and implemented an MIP-based optimization model using CPLEX, PL/SQL, and ASP.NET to allocate limited materials across product lines, converting BOM structures and business requirements into rigorous mathematical formulations • Improved material utilization and timely order fulfillment, cutting buyer workload by 95% and doubling the order success rate	
BEOL WPS Scheduling Optimization	09/2019 – 12/2019
• Replaced rule-based scheduling with a scalable MIP-based framework built with CPLEX, PL/SQL, and ASP.NET to allocate backend capacity across product lines under tight production constraints • Reduced WPS decommit rate by optimizing resource utilization, enhancing efficiency and overall supply chain responsiveness	

Publications [\[Full List\]](#)

[1] Chia-Yuan Wu, Frank E. Curtis, and Daniel P. Robinson. (2025). “A Bilevel Optimization Approach for Computing Synthetic Data to Mitigate Unfairness in Collaborative Machine Learning.” *Preprint in review.* [\[Paper\]](#)